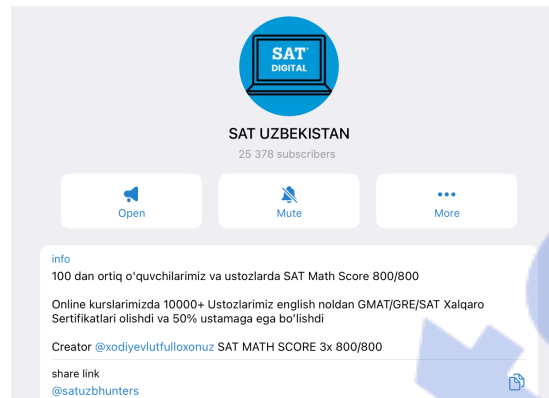
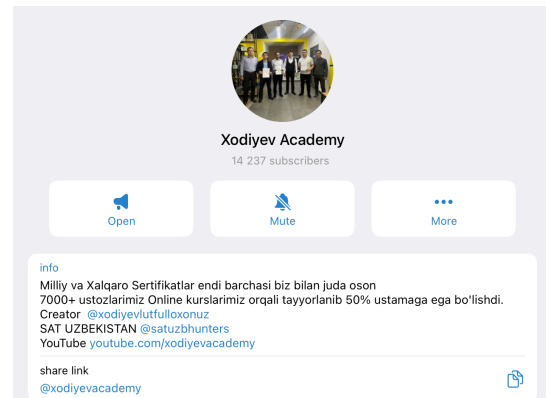


SAT UZBEKISTAN

Test creator **Xodiyev Lutfulloxon**
Telegram **@satuzbhunters**



Module-1

1.

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

2.

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{4\sqrt{2}}{9}$. What is the value of $\cos(S)$?

☐ (A) $\frac{4\sqrt{2}}{9}$

☒ (B) $\frac{4\sqrt{2}}{7}$

☐ (C) $\frac{7\sqrt{2}}{8}$

3.

Antwon's hair will grow at a constant rate from a length of 3 inches to a length of 5 inches. The equation $0.3t + 3 = 5$ describes this situation, where t is the number of months it will take for Antwon's hair to reach a length of 5 inches. Which of the following is the best interpretation of 0.3 in this context?

☐ (A) The time, in months, it will take Antwon's hair to grow 1 inch

☒ (B) The length, in inches, Antwon's hair will grow each month

☐ (C) The time, in months, it will take Antwon's hair to reach a length of 5 inches

4.

The product of a positive number x and the number that is 1 less than x is equal to 272. What is the value of x ?

5.

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 46$ in this context?

- ☒ A Approximately 46 grams of the sample remains 58 years after the sample starts to decay.
- ☐ B The mass of the sample has decreased by approximately 46 grams 58 years after the sample starts to decay.
- ☐ C The mass of the sample has decreased by approximately 58 grams 46 years after the sample starts to decay.
- ☐ D Approximately 58 grams of the sample remains 46 years after the sample starts to decay.

6.

Tanya drove on the highway and on local roads to complete a trip of 290 miles. The drive took 5 hours. She drove an average speed of 65 miles per hour (mph) on the highway and an average speed of 30 mph on local roads. If x is the time, in hours, Tanya drove on the highway and y is the time, in hours, she drove on local roads, which system of equations represents this situation?

- ☐ A $65x + 30y = 5$
 $x + y = 290$
- ☒ B $65x + 30y = 290$
 $x + y = 5$
- ☐ C $30x + 65y = 5$
 $x + y = 290$
- ☐ D $30x + 65y = 290$
 $x + y = 5$

7.

A dance studio charges an introductory fee for the first 3 lessons and then charges a fixed fee for each additional lesson. One student took 6 lessons and was charged \$150. Another student took 17 lessons and was charged \$480. Which function f gives the total charge, in dollars, for any student who took x lessons, where $x > 2$?

- ☐ A $f(x) = 25x$
- ☐ B $f(x) = 25x + 55$
- ☐ C $f(x) = 30x$
- ☒ D $f(x) = 30x - 30$

8.

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 23% every 11.69 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

☒ A $M = 100(0.77)^{\frac{t}{11.69}}$

☐ B $M = 100(0.77)^{t+11.69}$

☐ C $M = 100(0.23)^{\frac{t}{11.69}}$

☐ D $M = 100(0.23)^{t+11.69}$

9.

In triangle RST , the measure of angle R is 50° , the measure of angle S is x° , and the measure of angle T is $(4x - 5)^\circ$. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL ? (Disregard the degree symbol when entering your answer.)

10.

$$f(x) = \sqrt{6x + 7}$$

The function f is defined by the given equation. If $f(a) = -9a$, where a is a constant, what is the value of a ?

11.

Data set A: 8, 11, 12, 15, 21, 27

Data set B: 9, 12, 13, 16, 22, h

Data sets A and B each consist of 6 values as shown, where h is a constant. If the standard deviation of data set A is greater than the standard deviation of data set B, which of the following could be the value of h ?

I. 23

II. 28

III. 29

☒ A I only

☐ B II only

☐ C III only

☐ D II or III

12.

Luis has \$150 in an account. Each year he expects to have 1.1% more money in the account than he had the previous year. Which of the following models best describes how Luis expects the money in his account to change over time?

☐ (A) Decreasing exponential

☒ (B) Decreasing linear

☐ (C) Increasing exponential

13.

In right triangle ABC , angles A and B are acute, side AC has a length of 23.3, and $\tan B = \frac{1}{4}$. What is the length of side BC , rounded to the nearest tenth?

14.

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c . What percent of b is a ?

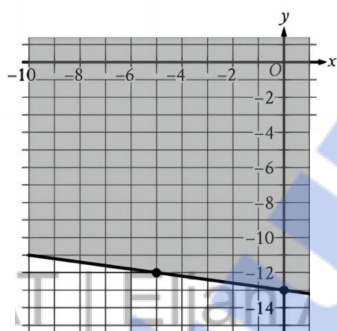
☐ (A) 4,706%

☒ (B) 2,600%

☐ (C) 38.12%

☐ (D) 21.87%

15.



The shaded region shown represents the solutions to $rx + ty \geq -65$, where r and t are constants. What is the value of $r + t$?

☐ (A) 6

☒ (B) 4

☐ (C) -4

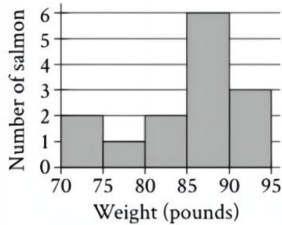
☐ (D) -5

16.

In the xy -plane, line k and line ℓ are perpendicular and intersect at the point $(7, 5)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line ℓ ?

- ☒ (A) $(8, 5 - \frac{1}{m})$
- ☐ (B) $(8, 5 + \frac{1}{m})$
- ☐ (C) $(8, 5 - m)$
- ☐ (D) $(8, 5 + m)$

17.

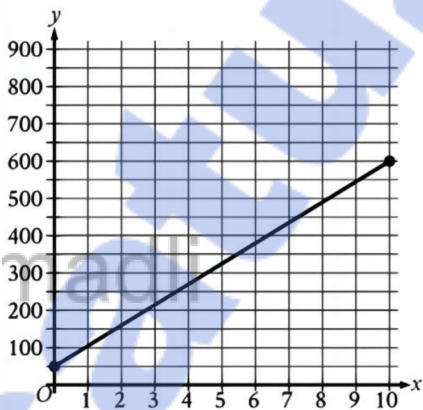


The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 32 pounds is added to the original data set to create a new data set of 15 weights, which of the following measures must be less for the new data set than for the original data set?

- I. The median
 - II. The mean
- ☐ (A) I only
 - ☐ (B) II only
 - ☒ (C) I and II
 - ☐ (D) Neither I nor II

18.

While walking on a path, Maksim stopped at a water fountain. The graph shows the total distance y , in meters, Maksim had walked on the path x minutes after leaving the water fountain.



What distance, in meters, did Maksim walk on the path in the 10 minutes after he left the water fountain?



19.

$$f(x) = 2x^2 + bx + c$$

The quadratic function $f(x)$ is defined above, where b and c are two constants. The function decreases when $x \leq 7$ and increases when $x \geq 7$. If the graph of the function touches the x -axis at two distinct points, what is the greatest possible value of $b + c$?

- A) 68
- B) 69
- C) 70
- D) 71

20.

The results of two independent surveys are shown in the table below.

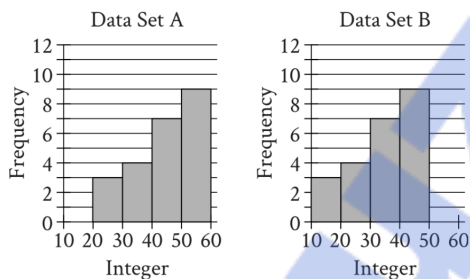
Men's Height

Group	Sample size	Mean (centimeters)	Standard deviation (centimeters)
A	2,500	186	12.5
B	2,500	186	19.1

Which statement is true based on the table?

- A. The Group A data set was identical to the Group B data set.
- B. Group B contained the tallest participant.
- C. The heights of the men in Group B had a larger spread than the heights of the men in Group A.
- D. The median height of Group B is larger than the median height of Group A.

21.



Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

(A) 0

(B) 1

(C) 10

(D) 23

22.

Poll Results

Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

(A) 163

(B) 1,304

(C) 3,864

(D) 5,621

SAT UZBEKISTAN

Test creator Xodiyev Lutfulloxon

Telegram @satuzbhunters

Module-2

1.

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(7, -6)$. Which of the following points also lies on line t ?

(A) $(0, \frac{8}{7})$

(B) $(6, 9)$

(C) $(14, 2)$

(D) $(15, 1)$

2.

In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 24 and the length of $\overline{AC}$ is 13 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

(A) $\frac{13}{24}$

(B) $\frac{24}{37}$

(C) $\frac{37}{24}$

(D) $\frac{24}{13}$

3.

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 56 square centimeters (cm^2), and the surface area of right rectangular prism Y is 896 cm^2 . The volume of right rectangular prism Y is 1,536 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3 , of right rectangular prism X and right rectangular prism Y?



4.

$$kx^2 - 16x = 24x^2 - 8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

5.

The function f is defined by $f(x) = \frac{x^2+ax+b}{2x+c}$, where a , b , and c are constants. The graph of the function f in the xy -plane, where $y = f(x)$, does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

6.

To investigate the effect of magnesium supplementation on the sleep quality of adults, a researcher selected 190 adults at random from a community center to participate in a study. The researcher first measured the sleep efficiency of each participant. Each participant was then randomly assigned to take a magnesium supplement or a placebo each day for 6 weeks. At the end of the 6 weeks, the researcher measured the sleep efficiency of all participants again and found that the magnesium supplement caused statistically significant improved sleep quality for the adults at the community center. What feature of this study allowed the researcher to conclude that the magnesium supplement caused the improved sleep quality?

(A) There were more than 100 participants in the study.

(B) The participants were selected at random from the community center.

(C) The sleep efficiency of each participant was measured again at the end of 6 weeks.

(D) Each participant was randomly assigned to take a magnesium supplement or a placebo.

7.

For the function f , for every increase of 2 in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant. Which of the following equivalent forms of function f displays the value of c as the base or the coefficient?

(A) $f(x) = 42(2)^{6x}$

(B) $f(x) = 42(8)^{2x}$

(C) $f(x) = 42(64)^x$

(D) $f(x) = 42(4,096)^{\frac{x}{2}}$

8.

$$125x^2 - (25a + 5b)x + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(5a + b)$, where k is a constant. What is the value of k ?

9.

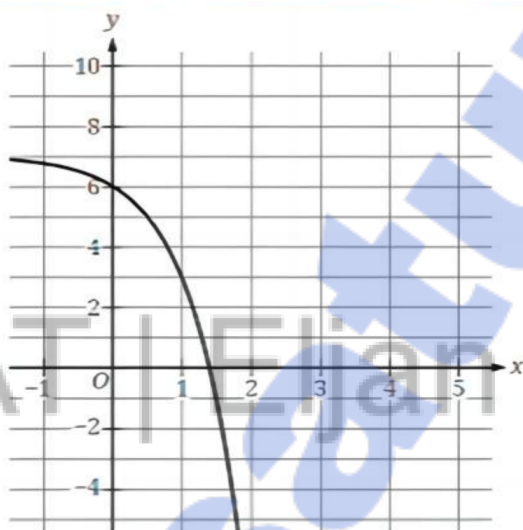
The function f is defined by $f(x) = \frac{|x|}{a} - 16$, where $a < 0$. What is the product of $f(17a)$ and $f(8a)$?

10.

A circle in the xy -plane has its center at $(-2, 2)$. Line t is tangent to this circle at the point $(3, -1)$. Which of the following points also lies on line t ?

☐ (A) $(0, \frac{5}{3})$
☐ (B) $(1, 7)$
☐ (C) $(6, 4)$
☐ (D) $(8, 2)$

11.



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?

☐ (A) $f(x) = -4^x + 3$
☐ (B) $f(x) = -4^x + 5$
☐ (C) $f(x) = -4^x + 6$
☐ (D) $f(x) = -4^x + 7$

12.

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 10 and 12, which inequality represents the possible lengths, x , of the third side of the triangle?

(A) $x < 22$

(B) $x > 22$

(C) $2 < x < 22$

(D) $x < 2$ or $x > 22$

13.

A painting studio is offering a promotion where the first class is free, the second class is half off the regular price, and the remaining classes are regularly priced. If the regular price of a class is \$26.40, which function f gives the total cost, in dollars, of x classes taken using this promotion, where $x \geq 2$?

(A) $f(x) = 26.40(x - 1) + 13.20$

(B) $f(x) = 26.40(x - 2) + 13.20$

(C) $f(x) = 26.40(x - 1) + 13.20(x - 2)$

(D) $f(x) = 26.40(x - 2) + 13.20(x - 1)$

14.

A clothing store buys shirts at a wholesale price of 6.00 dollars each and resells them each at a retail price that is 310% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 75% off the retail price. What is the discounted price of each remaining shirt, in dollars?

15.

The positive number a is 2,184% of the sum of the positive numbers b and c , and b is 84% of c . What percent of b is a ?

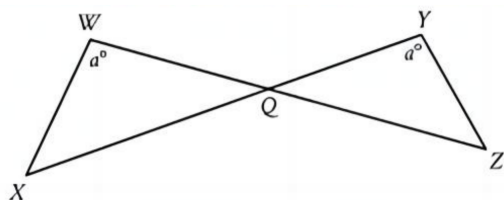
(A) 4,784%

(B) 2,600%

(C) 40.19%

(D) 22.68%

16.



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 27$, $WQ = 90$, $WX = 60$, and $XQ = 110$. What is the length of YZ ?

(A) 18

(B) 33

(C) 110

(D) 200

17.

$$x(r - 7) + 3 = 17x + 25$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

(A) 3

(B) 7

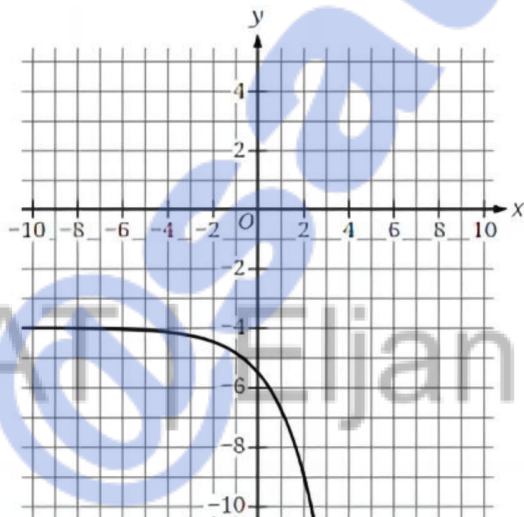
(C) 22

(D) 24

18.

The equation $h = \frac{9(v - 273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of 809.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

19.



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

(A) Three

(B) Two

(C) One

(D) Zero

20.

An object is launched from a height of 124 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 3 seconds after the object is launched, it reaches a maximum height of 268 feet above the ground. Which equation defines f ?

(A) $f(t) = -16(t + 268)^2 - 3$

(B) $f(t) = -16(t - 268)^2 + 3$

(C) $f(t) = -16(t + 3)^2 - 268$

(D) $f(t) = -16(t - 3)^2 + 268$

21.

In one week in 2017, a technician earned a total of \$900 by working at her regular job and at a second job doing part-time work. The equation $21h + 10c = 900$ represents this situation, where h is the number of hours worked at her regular job and c is the number of hours worked at her second job. Which of the following is the best interpretation of 21 in this context?

(A) The amount, in dollars, the technician earned for each hour she worked at her regular job

(B) The amount, in dollars, the technician earned for each hour she worked at her second job

(C) The number of hours the technician worked in one week at her regular job

(D) The number of hours the technician worked in one week at her second job

22.

A sphere has a volume of 245.00 cubic yards. What is the volume of the sphere, in cubic meters? (Use 1 yard = 0.914 meter.)